

# THE INTERNET OF THINGS



News, views & development on the Internet of Things

## What is the IoT?

The answer, in plain English...

Some think the Internet of Things (IoT) is a world where everything is digital, where we will no longer be alone, where machines talk more than we do, and everything is intelligent. While this might happen decades down the line, the fact is that the IoT need not necessarily be so disruptive. It could be evolutionary, and that is the practical approach that many technologists are taking today. That is the logic behind the many hardware and software innovations that claim to be a part of the IoT movement.

### A more practical approach

There is a more practical majority that looks at the Internet of Things as a layer of digital technology added atop the objects in this world—smartphones, consumer goods, home appliances, governance tools, office infrastructure, health support and just about everything. This digital layer is woven together by a smart fabric (say, the World Wide Web and its apps) that adds meaning to this digital layer.

There are several technologies that make this possible today. There are chips or codes on the objects. These store information at the most basic level—for example, bar codes or QR codes. More advanced technologies also include local-area communication capabilities—for example, radio frequency identification (RFID), near-field communication (NFC) and Zigbee-capable chips. Even more advanced technologies have embedded systems.

These objects store and sometimes also process information, and communicate with other devices. This enables the information on the chip to be transferred to a larger application, as well as to be updated. At the next layer, there are applications that convert this information into something useful. The infor-

mation might be used to provide a customised service to the user, to configure the object, to track its movement or for statistical purposes.

### IoT at work

Although the full potential of the IoT is yet to be exploited, it already manifests in many forms around us.

A QR code may be printed on a building façade. When you click this with your phone, you get to see a list of offices or shops inside the building.

Smart thermostats (Nest) and switches (Belkin WeMo) enable day-to-day appliances to be controlled with your phone.

RFID labels on pets, people, trucks, containers, pallets and goods enable their movement to be monitored easily.

Health monitoring equipment, such as blood glucose monitors from Abbott and Medtronic; fitness and heart rate monitors from makers like Garmin, Adidas, Polar and Suunto; as well as fitness tools like Nike's Fuelband, Larklife and Fitbit, connect to smartphones and enable self-help or remote monitoring of patients by doctors.

There are cameras embedded on traffic signals, pavements and medians that spot over-speeding and levy appropriate fines.

Chips on products are used to notify shoppers about offers and complementary products, and smartphones are used to pay for goods.

All these are simple examples of the IoT at work—a layer of technology wrapping common objects woven together by a network; and great apps that utilise this infrastructure to deliver some benefit to users.

Now you decide—has the IoT era begun or not?

### Smart City



As one of the most extensive examples of the IoT, City24/7—along with Cisco IBSG and LG Electronics—has launched an interactive platform that integrates information from open government programmes, local businesses and citizens to provide meaningful and powerful knowledge anytime, anywhere, on any device. Currently operational in New York, City24/7 uses touch-, voice- and audio-capable smart screens located at crowded public places in the city to deliver a wide array of real-time, hyper-local (about two square city blocks) information and services. City24/7 can also be accessed on smartphones, tablets and laptops by connecting to the smart screens using Wi-Fi. The goal of the scalable solution is to deliver the information people need to know, where and when they need it.

### IoT-style toys



Atoms Express is a plug-and-play electronics construction kit from Seamless Toy Company that lets you make low-cost connected toys. Without any prior programming or engineering experience, you can use the colour-coded blocks, sensors, components (like motors and lights) and the Bluetooth Smart Brick to put together smart toys that can interact with other toys and be controlled with any iOS device.